Regression Assignment – Insurance Charges Prediction

1. Problem Statement

The objective of this assignment is to build a machine learning regression model that predicts insurance charges based on several input parameters such as age, BMI, number of children, gender, and smoking status.

The dataset provided by the client contains multiple attributes related to insurance policy holders. Using this dataset, different regression models are trained and evaluated to find the best model for predicting insurance charges.

- **Multiple Linear Regression $R^{2\ value}$ is = 0.7894**

- **Dataset Information**
  Total Rows: 1338
  Total Columns: 6

2. Support Vector Machine (SVM Regression)

Different hyperparameter values were tested using multiple kernels to evaluate the model performance.

S.NO	Hyper Parameter	Linear (R^2)	RBF (R^2)
1	C10	0.4320	0.2105
2	C20	0.5124	0.3507
3	C50	0.5986	0.5208
4	C100	0.6421	0.6304
5	C500	0.7015	0.7423
6	C1000	0.7452	**0.8125**

The SVM regression used best R^2 value **= 0.81**

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4.Decision Tree Regression

Different combinations of Criterion, Max_Depth, and Splitter parameters were tested

S.No	Criterion	Splitter	Max_Depth	R_Value
1	squard_error	best	None	0.68
2	squard_error	Best	5	0.83
3	squard_error	Random	5	0.86
4	Squard_error	Random	10	0.79
5	Squard_error	Best	15	0.68
6	Squard_error	Random	None	0.69
7	Friedman_mse	Best	None	0.70
8	Friedman_mse	Best	5	0.84
9	Friedman_mse	Random	10	0.72
10	Friedman_mse	Best	20	0.68
11	Absolute_error	Best	5	0.86
12	Absolute_error	Random	10	0.83
13	Absolute_error	Best	3	0.87
14	Poisson	Best	5	0.84
15	Poisson	Random	10	0.82

The Decision Tree Regression used best R^2 value **(Absolute_ error , 3, Best)** = 0.87

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5. Random Forest Regression

Random Forest regression models were trained using different numbers of estimators and feature selections.

Model No	n_estimators	Max_depth	Min_samples_split	Min_samples_leaf	Max_features	R^2 Score
RF-01	50	None	2	1	None	0.84
RF-02	100	None	2	1	None	0.85
RF-03	200	None	2	1	None	0.85
RF-04	300	10	2	1	None	0.85
RF-05	200	15	2	1	None	0.85
RF-06	500	20	2	1	None	0.85
RF-07	300	None	4	1	None	0.86
RF-08	300	None	2	2	None	0.88
RF-09	400	None	5	2	None	0.87
RF-10	500	25	4	1	None	0.86
RF-11	600	None	2	1	None	0.85
RF-12	700	None	2	1	None	0.85
RF-13	800	30	2	1	1.0	0.85
RF-14	900	35	3	1	1.0	0.85
RF-15	1000	None	2	1	1.0	0.85

RF-08 and 09 achieved the highest performance with an **R^2 score =0.88**, indicating the best predictive capability

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6. Final Model Selection

Multiple regression models such as Support Vector Machine, Decision Tree, and Random Forest were tested and evaluated using the R^2 score metric.

Among all the models, the Random Forest Regression model achieved the highest

R^2 score of 0.88. Therefore, Random Forest was selected as the final model for predicting insurance charges.